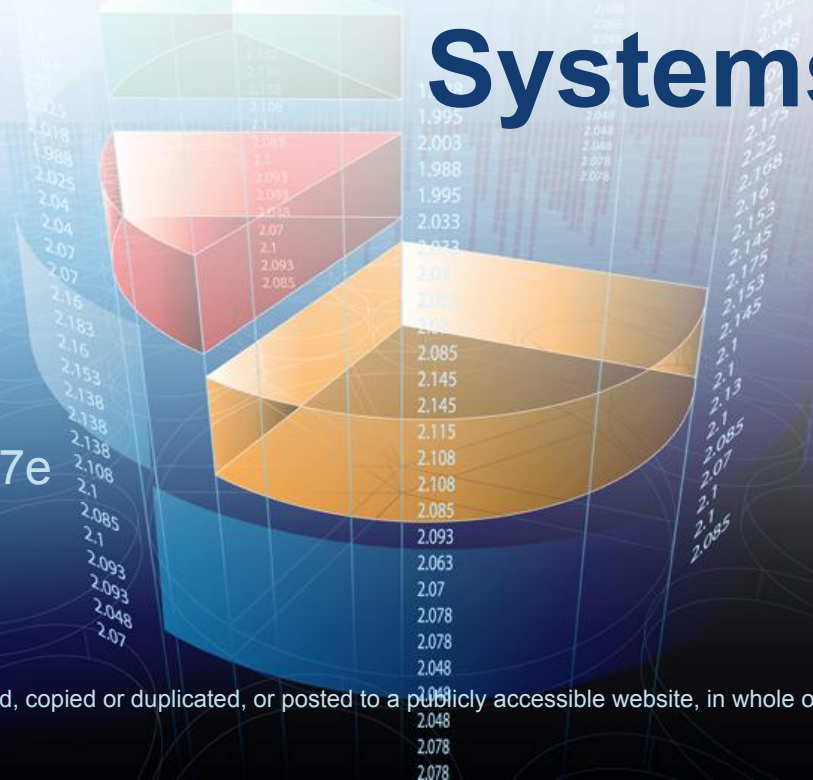
Chapter 8
Financial Reporting and Management Reporting Systems



Accounting Information Systems, 7e
James A. Hall

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Objectives for Chapter 8

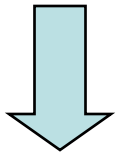
- Understand the operational features of the General Ledger System(GLS), financial reporting system(FRS), and management reporting system(MRS).
- Be able to identify the principle operational controls governing the GLS and FRS.
- Understand the factors that influence the design of the MRS.
- Understand the elements of a responsibility accounting system.
- Be familiar with the financial reporting issues surrounding XBRL.

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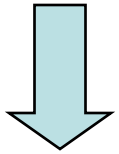
IS Functions of GLS

- General ledger systems should:
  - collect transaction data promptly and accurately.
  - classify/code data and accounts.
  - validate collected transactions/ maintain accounting controls (e.g., equal debits and credits).
  - process transaction data.
    - post transactions to proper accounts
    - update general ledger accounts and transaction files
    - record adjustments to accounts
  - store transaction data.
  - generate timely financial reports.

Input



Process



Output

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Relationship of GLS to Other Information Subsystems

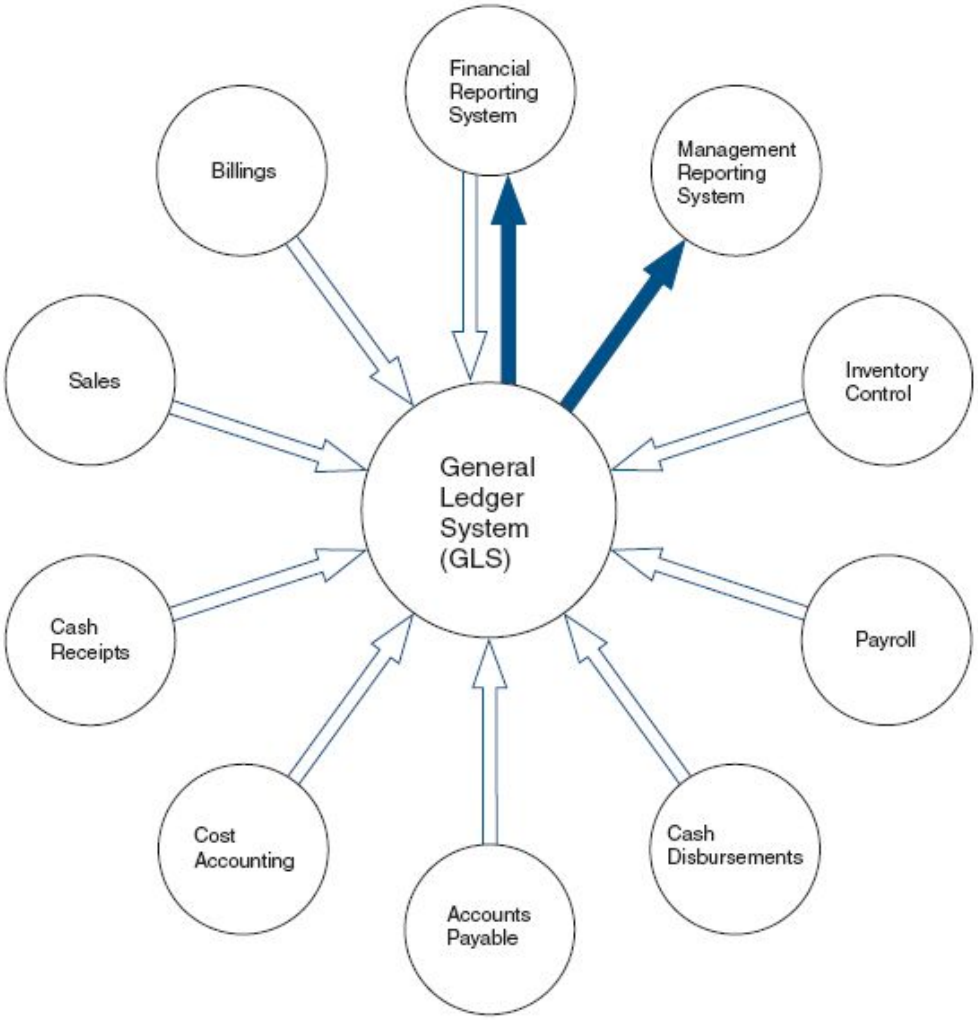


Figure 8-1

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GLS Database

- General ledger master file
  - *principal FRS file based on chart of accounts*
- General ledger history file
  - *used for comparative financial support*
- Journal voucher file
  - *all journal vouchers of the current period*
- Journal voucher history file
  - *journal vouchers of past periods for audit trail*
- Responsibility center file
  - *financial data by responsibility centers for MRS*
- Budget master file
  - *budget data by responsibility centers for MRS*

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Journal Voucher Layout for a General Ledger Master File

Journal Voucher

Number: JV6 - 03
Date: 6/26/04

Acct Num	Account Name	Amount DR.	Amount CR.
130	Accts Rec.	$5,500	
502	Sales		$5,500

Explanation: To Record Total Credit Sales for 6/26/04.

Approved By: J. A. Martin

Posted By: S. D. Smith

Figure 8-2

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Financial Reporting Process

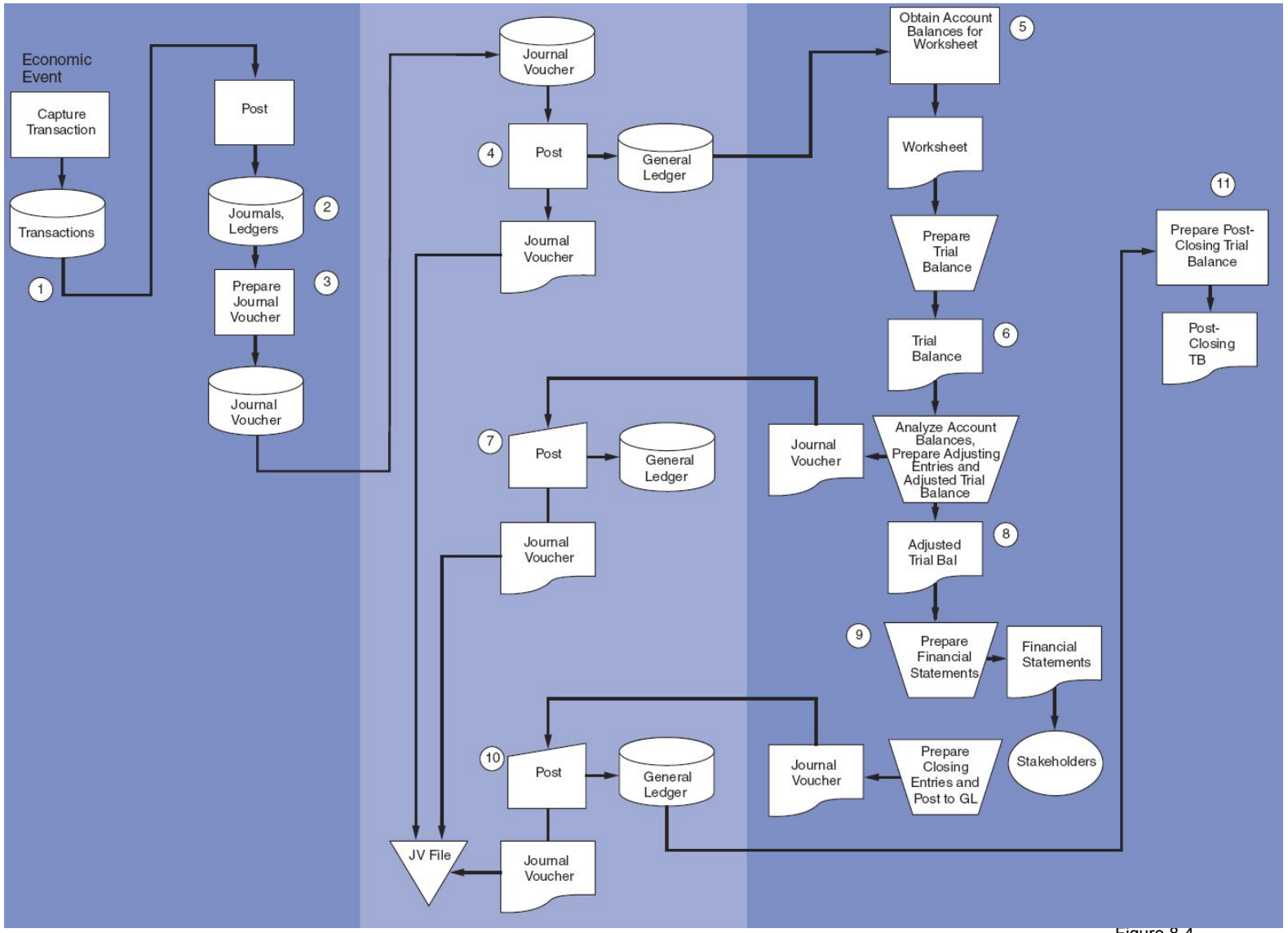


Figure 8-4

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GLS Reports

- General ledger analysis:
  - listing of transactions
  - allocation of expenses to cost centers
  - comparison of account balances from prior periods
  - trial balances
- Financial statements:
  - balance sheet
  - income statement
  - statement of cash flows
- Managerial reports:
  - analysis of sales
  - analysis of cash
  - analysis of receivables
- Chart of accounts: coded listing of accounts

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Potential Risks in the GL/FRS

- Improperly prepared journal entries
- Unposted journal entries
- Debits not equal to credits
- Subsidiary not equal to G/L control accounts
- Inappropriate access to the G/L
- Poor audit trail
- Lost or damaged data
- Account balances that are wrong because of unauthorized or incorrect journal vouchers

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GL/FRS Control Issues

- **Transaction authorization** - journal vouchers must be authorized by a manager at the source dept
- **Segregation of duties** – G/L clerks should not:
  - have recordkeeping responsibility for special journals or subsidiary ledgers
  - prepare journal vouchers
  - have custody of physical assets

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GL/FRS Control Issues

- **Access controls:**
  - Unauthorized access to G/L can result in errors, fraud, and misrepresentations in financial statements.
  - Sarbanes-Oxley requires controls that limit database access to only authorized individuals.
- **Accounting records** - trace source documents from inception to financial statements and vice versa

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GL/FRS Control Issues

- **Independent verification**
  - G/L dept. reconciles journal vouchers and summaries.
- Two important operational reports used:
  - *journal voucher listing* – details of each journal voucher posted to the G/L
  - *general ledger change report* – the effects of journal voucher postings on G/L accounts

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GL/FRS Using Database Technology

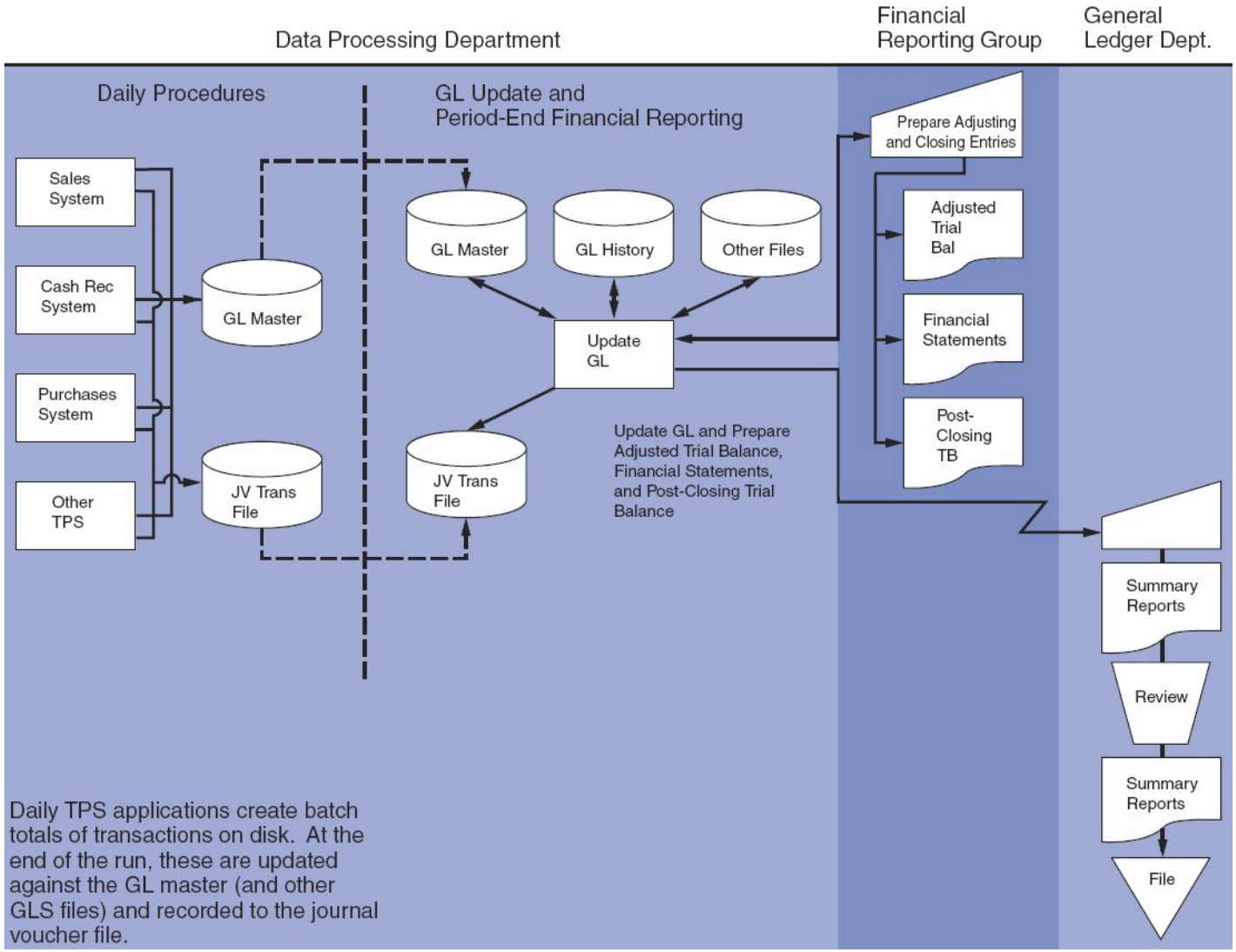


Figure 8-5

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GL/FRS Using Database Technology

- Advantages:
  - immediate update and reconciliation
  - timely, if not real-time, information
- Removes separation of transaction authorization and processing
  - Detailed journal voucher listing and account activity reports are a compensating control
- Centralized access to accounting records
  - Passwords and authorization tables as controls

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HTML: Hyper Text Markup Language

- Format used to produce Web pages
  - defines the page layout, fonts, and graphic elements
  - used to lay out information for display in an appealing manner like one sees in magazines and newspapers
  - using both text and graphics (including pictures) appeals to users
- Hypertext links to other documents on the Web
  - Even more pertinent is HTML's support for hypertext links in text and graphics that enable the reader to 'jump' to another document located anywhere on the World Wide Web.

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XML: eXtensible Markup Language

- XML is a meta-language for describing markup languages.
- *Extensible* means that any markup language can be created using XML.
  - includes the creation of markup languages capable of storing data in relational form, where tags (formatting commands) are mapped to data values
  - can be used to model the data structure of an organization's internal database

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Comparison of HTML and XML Documents

Partial HTML Book Order

```
<H1>Book Order</H1>
<BOLD>Understanding XML</BOLD>
<H2>Doe, John</H2>
<ITALIC>1</ITALIC>
<BOLD>9.95</BOLD>
<H2>Standard UPS </H2>
```

End-user computer cannot process HTML and can only display the format.



Partial XML Book Order

```
<ORDERTYPE>Book Order</ORDERTYPE>
<TITLE>Understanding XML</TITLE>
<AUTHOR>Doe, John</AUTHOR>
<QUANTITY>1</QUANTITY>
<PRICE>9.95</PRICE>
<SHIPPING>Standard UPS </SHIPPING>
```

End-user computer can recognize XML and process accordingly, relieving some of the burden currently placed on web servers.



Figure 8-6

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XBRL: eXtensible Business Reporting Language

- **XBRL** is an XML-based language for standardizing methods for preparing, publishing, and exchanging financial information, e.g., financial statements.
- **XBRL taxonomies** are classification schemes.
- Advantages:
  - Business offer expanded financial information to all interested parties virtually instantaneously.
  - Companies that use XBRL database technology can further speed the process of reporting.
  - Consumers import XBRL documents into internal databases and analysis tools to greatly facilitate their decision-making processes.

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Implications for Accounting

- Audit implication for XBRL
  - ***taxonomy creation:*** incorrect taxonomy results in invalid mapping that may cause material misrepresentation of financial data
  - ***validation of instance documents:*** ensure that appropriate taxonomy and tags have been applied
  - ***audit scope and timeframe:*** impact on auditor responsibility as a consequence of real-time distribution of financial statements

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Management Reporting Systems

- Produce financial and nonfinancial information needed by management to "plan, evaluate, control"
- Usually seen as discretionary reporting
- Can argue that Sarbanes-Oxley requires MRS
  - MRS provide a formal means for monitoring the internal controls

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Factors That Influence MRS Design

- Management principles
- Management function, level, and decision type
- Problem structure
- Types of management reports
- Responsibility accounting
- Behavioral considerations

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Management Principles

- Formalization of tasks:
  - structures the firm around the tasks performed rather than around individuals' unique skills
  - allows specification of the information needed to support the tasks

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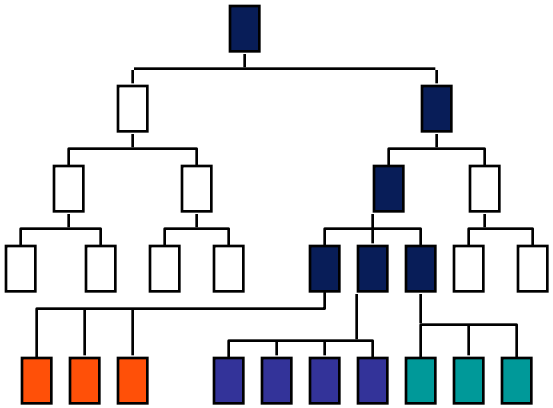
Management Principles

- Responsibility and authority:
  - *responsibility* - obligation to achieve desired results
  - *authority* - power to make decisions within the limits of that responsibility
  - delegated by managers to subordinates
  - define the vertical reporting channels through which information flows

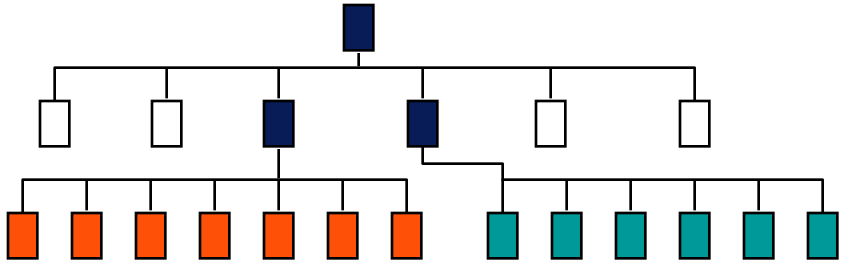
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Management Principles

- **Span of control:**
  - the number of subordinates directly under the manager's control
  - detailed reports for managers with narrow spans of control
  - summarized information for managers with broad spans of control



Narrow Span of Control



Wide Span of Control

Figure 8-15

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Management Principles

- Management by exception:
  - Managers should limit their attention to potential problem areas.
  - Reports should focus on changes in key factors that are symptomatic of potential problems.

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Management Level and Decision Type



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Management Function, Level, and Decision Type

- Strategic planning decisions:
  - firm's goals and objectives
  - scope of business activities
  - organizational structure
  - management philosophy
  - long-term, with broad scope and impact
  - non-recurring , with high degree of uncertainty
  - need highly summarized information
  - require external & internal information sources

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Management Function, Level, and Decision Type

- Tactical planning decisions:
  - subordinate to strategic decisions
  - short term
  - specific objectives
  - recur often
  - fairly certain outcomes
  - limited impact on the firm

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Management Function, Level, and Decision Type

- Management control decisions:
  - using resources as productively as possible in all functional areas
  - evaluating the performance of subordinates against standards
- Measuring performance is difficult because sound decisions with long-term benefits may negatively impact the short- term bottom line.

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Management Function, Level, and Decision Type

- Operational control decisions:
  - deal with routine tasks
  - narrower focus, dependent on details
  - highly structured
  - short time frame
- Three basic elements or steps:
  - set attainable standards
  - evaluate performance
  - take corrective action

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Classification of Decision Types by Decision Characteristics

	DECISION TYPE			
DECISION CHARACTERISTIC	**Strategic Planning**	**Tactical Planning**	**Management Control**	**Operational Control**
Time frame	Long term	Medium	Medium	Short
Scope	High impact	Medium impact	Lower impact	Lowest impact
Level of details	Highly summarized	Detailed	Moderately summarized	Highly detailed
Recurrence	Nonrecurring	Periodic recurring	Periodic recurring	Frequent recurring
Certainty	Uncertain	Highly certain	Uncertain	Highly certain

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Problem Structure

- Reflects and affects how well decision makers understand and solve problems
- Elements of problem structure:
  - data
  - procedures
  - objectives

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Problem Structure

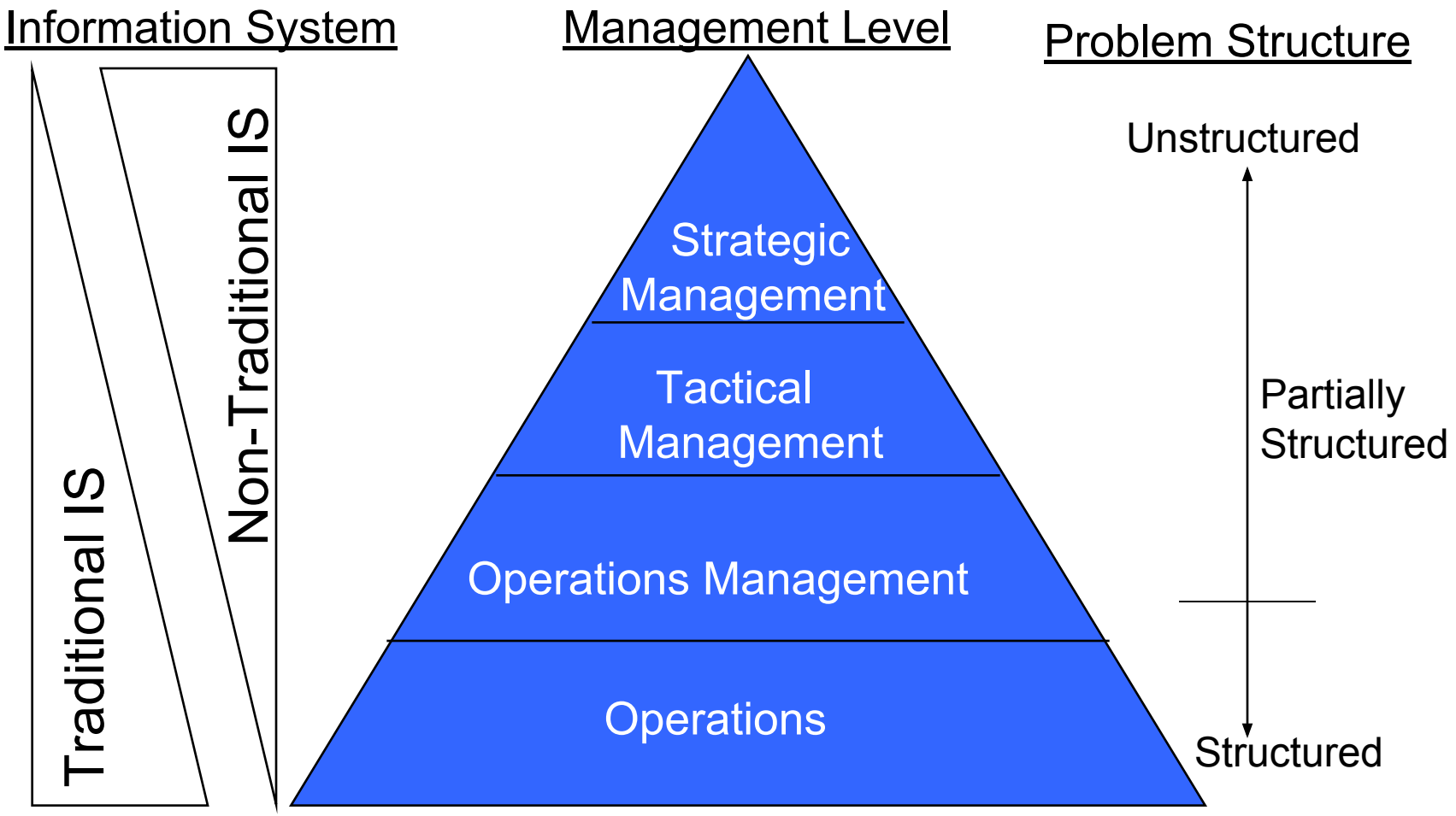


Figure 8-17

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Management Reports

- Report objectives - reports must have value or information content
- They should…
  - reduce the level of uncertainty associated with a problem facing the decision maker
  - influence the behavior of the decision maker in a positive way

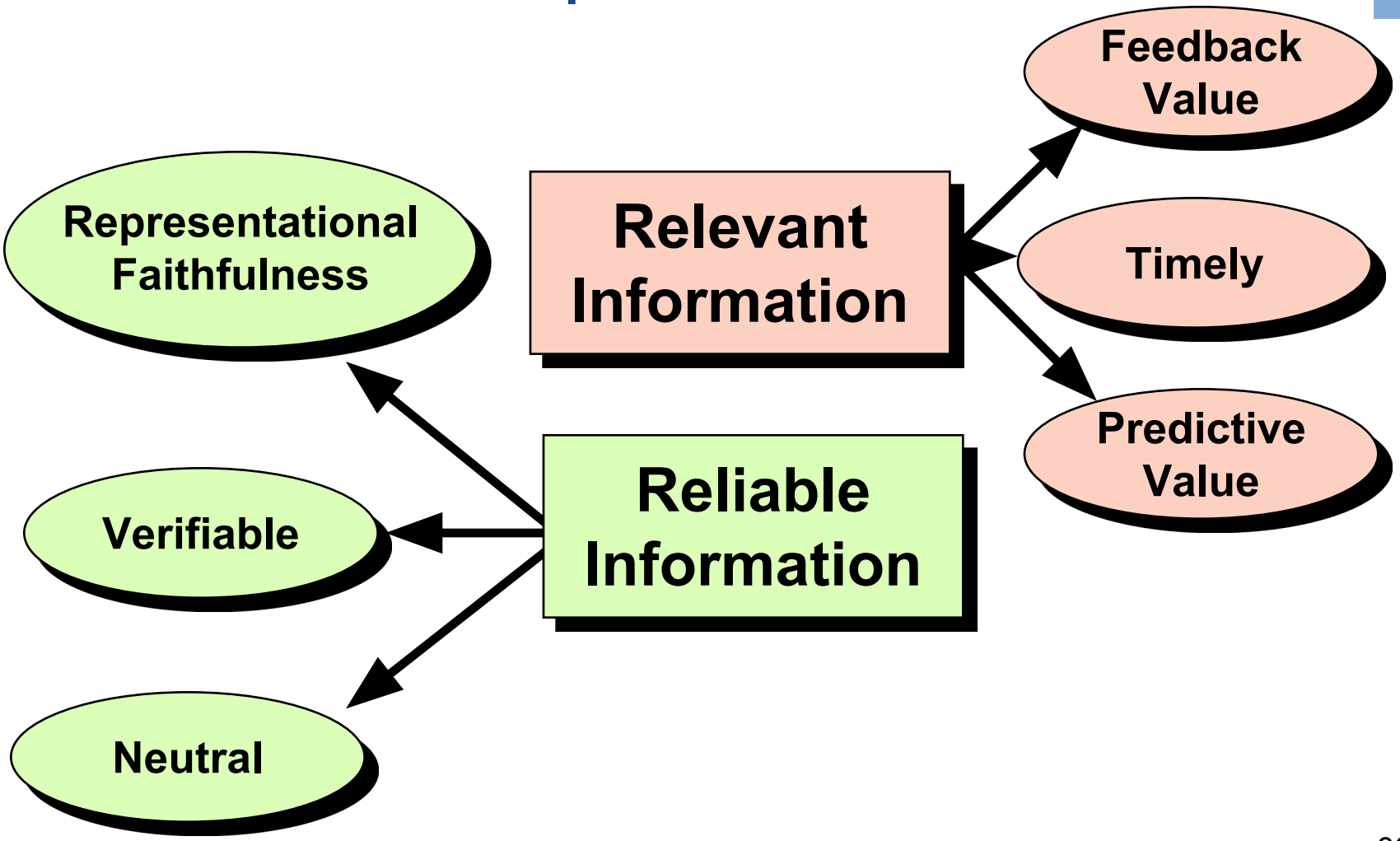
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Report Attributes

- Relevance – useful to decision making
- Summarization – appropriate level of detail
- Exception orientation – identify risks
- Accuracy – free of material errors
- Completeness – essential information
- Timeliness – in time for decisions
- Conciseness – understandable format

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Attributes of Useful Information According to FASB's Conceptual Framework



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Types of Management Reports

- Programmed reports:
  - scheduled reports – produced at specified intervals, e.g., weekly
  - on-demand reports – triggered by events, e.g., inventory levels drop to a certain level
- Ad hoc reports:
  - designed and created “as needed”
  - situations arise that require new information

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Responsibility Accounting

- Implies that every economic event that affects the organization is the responsibility of and can be traced to an individual manager
- Incorporates the fundamental principle that responsibility-area managers are accountable for items that they control

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Setting Financial Goals: Budgeting

- Budgeting helps management achieve financial objectives by setting measurable goals for each organizational segment.
- Budget information flows downward and becomes increasingly detailed at each lower level.
- The performance information flows upward as responsibility reports.

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Responsibility Centers

- *Cost center* – responsible for keeping costs within budgetary limits
- *Profit center* – responsible for both cost control and revenue generation
- *Investment center* – has general authority to make a wide range of decisions affecting costs, revenue, and investments in assets

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Behavioral Considerations: Goal Congruence

- MRS and compensation schemes help to appropriately assign authority and responsibility.
- If compensation measures are not carefully designed, managers may engage in actions not optimal for the organization.
  - Short-term v. long-term measures

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Behavioral Considerations: Information Overload

- Occurs when managers receive more information than they can assimilate.
- Can cause managers to disregard formal information and rely on informal—probably inferior—cues when making decisions.

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Behavioral Considerations: Performance Measures

- Appropriate performance measures
  - Stimulate behavior consistent with firm objectives.
  - Managers consider all relevant aspects, not just one.
- Example of inappropriate measures:
  - price variance – can affect the quality of the items purchased
  - quotas – can affect quality control, material usage efficiency, labor relations, plant maintenance
  - profit measures – can affect plant investment, employee training, inventory reserve levels, customer satisfaction

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